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(54) **ELECTRONIC CIGARETTE WITH
OPTIMISED VAPORISATION**

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None

See application file for complete search history.

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Primary Examiner — Felix O Figueroa

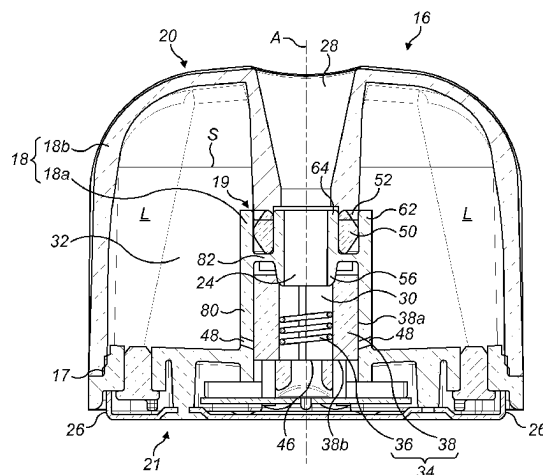
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(57)

ABSTRACT

A capsule for an electronic cigarette has a first end for
engaging with an electronic cigarette device and a second
end configured as a mouthpiece portion having a vapor
outlet. The capsule further includes a liquid store configured
to contain a liquid to be vaporized, a vaporizing unit
comprising a heater and a fluid transfer element, the vapor-
izing unit being arranged within a vaporizing chamber, a
main vapor channel extending from the vaporizing chamber
to the vapor outlet in the mouthpiece, and a housing enclos-
ing the liquid store and the vaporizing unit, wherein the fluid
transfer element is fluidly connected to the liquid store by at
least one liquid inlet and the fluid transfer element provides
a capillary action on liquid received therein, wherein the
heater is provided at a position that is substantially adjacent

(Continued)



the liquid inlet, or at a position between the liquid inlet and the mouthpiece.

20 Claims, 6 Drawing Sheets

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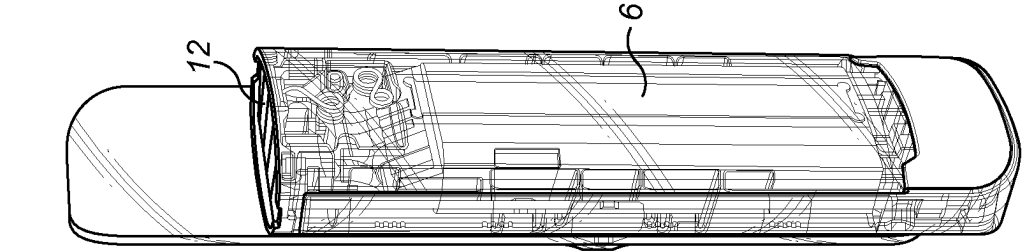


FIG. 1c

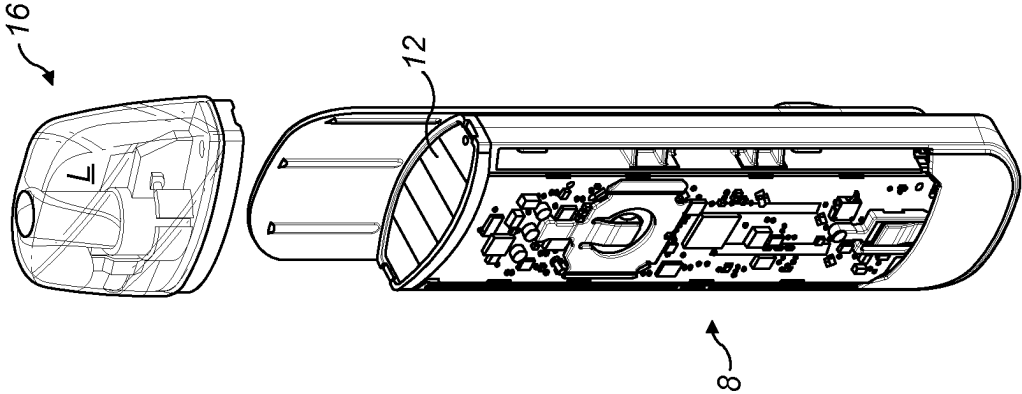


FIG. 1b

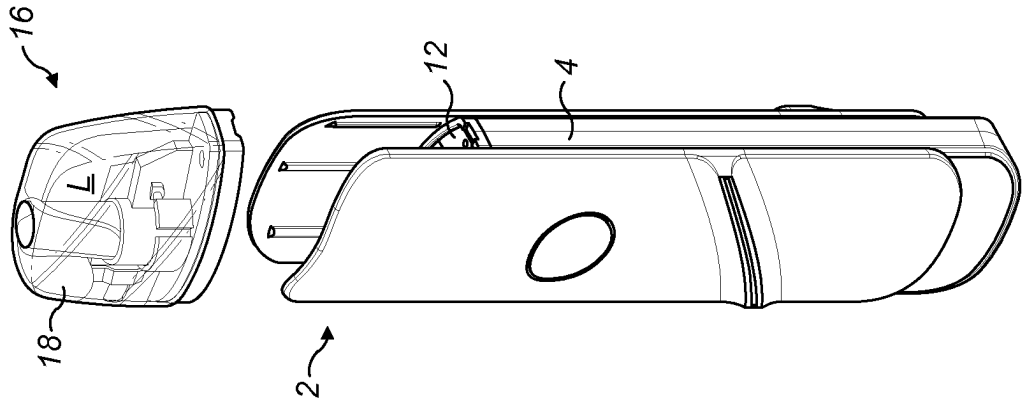


FIG. 1a

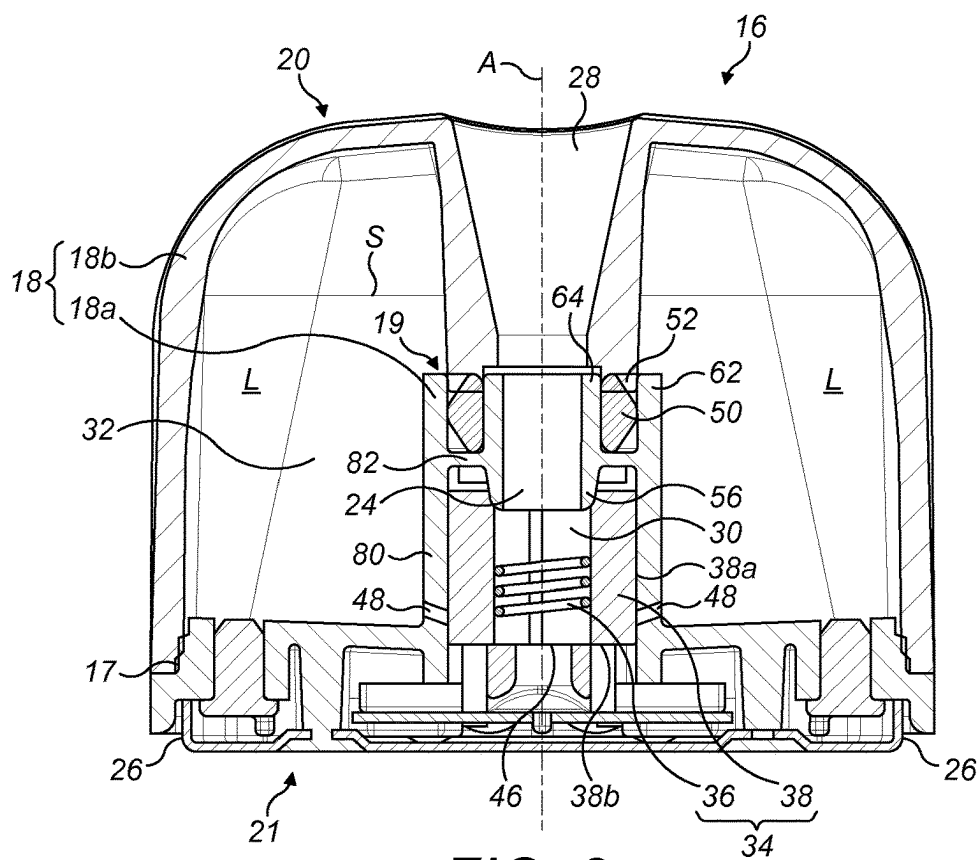


FIG. 2a

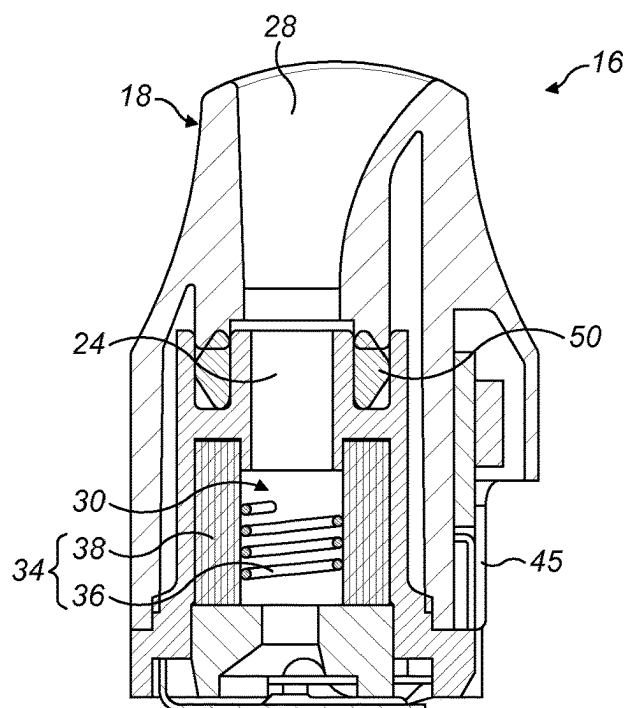


FIG. 2b

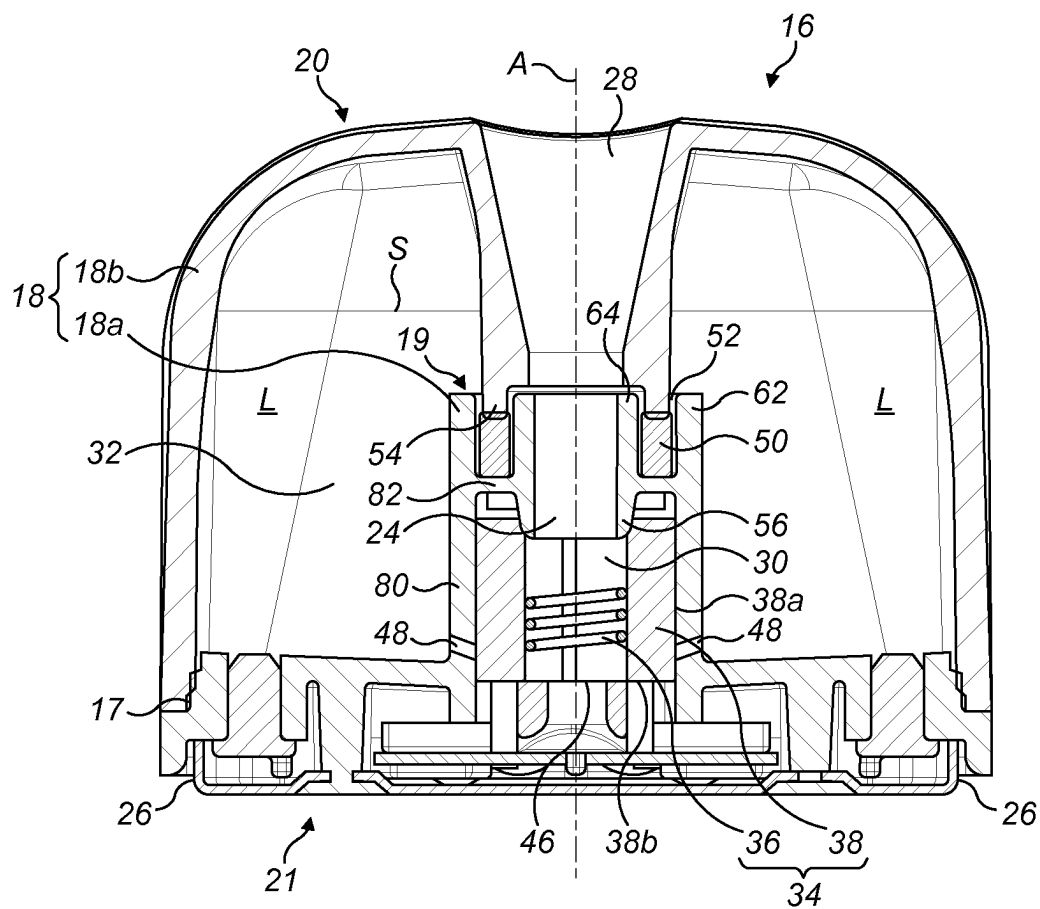


FIG. 2c

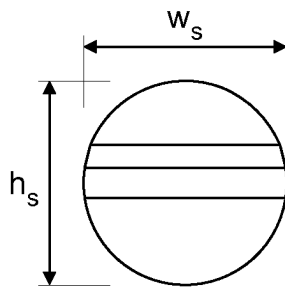


FIG. 3a

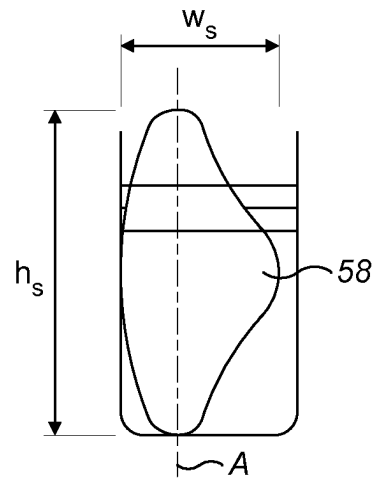


FIG. 3b

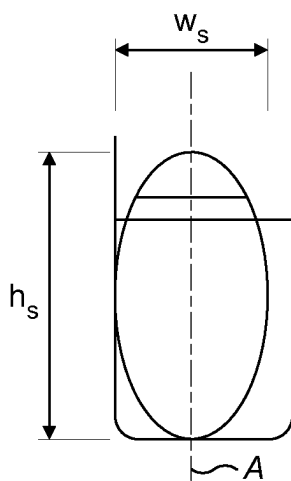


FIG. 3c

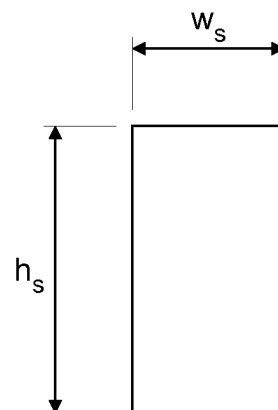


FIG. 3d

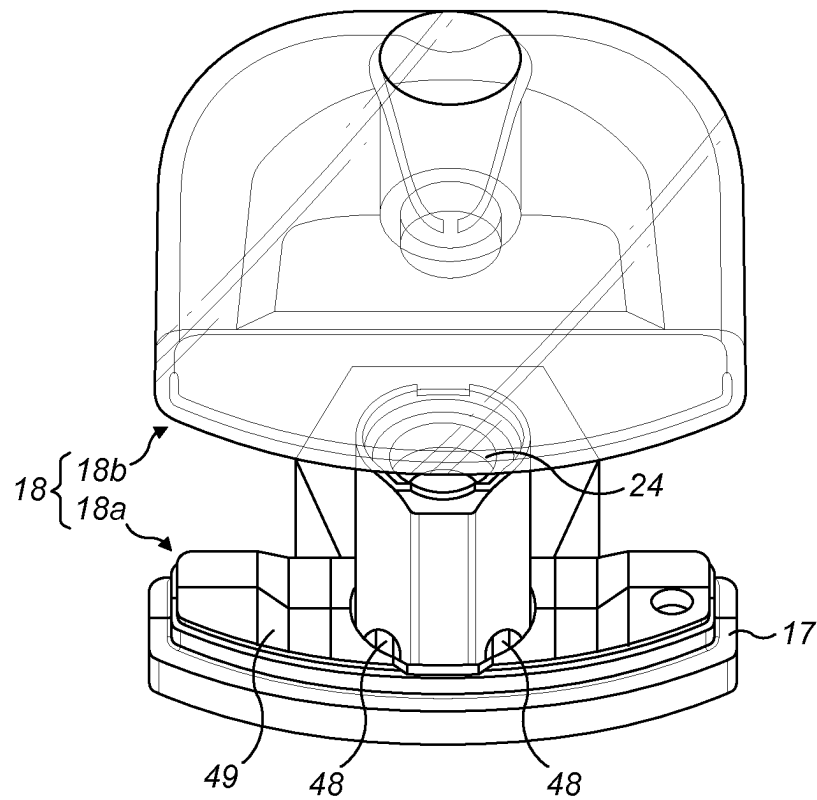


FIG. 4a

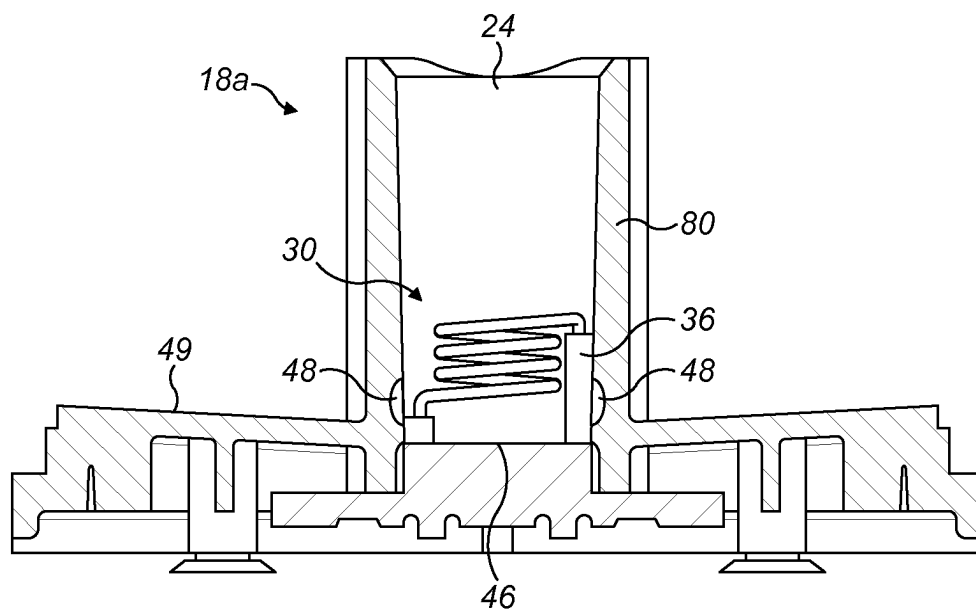


FIG. 4b

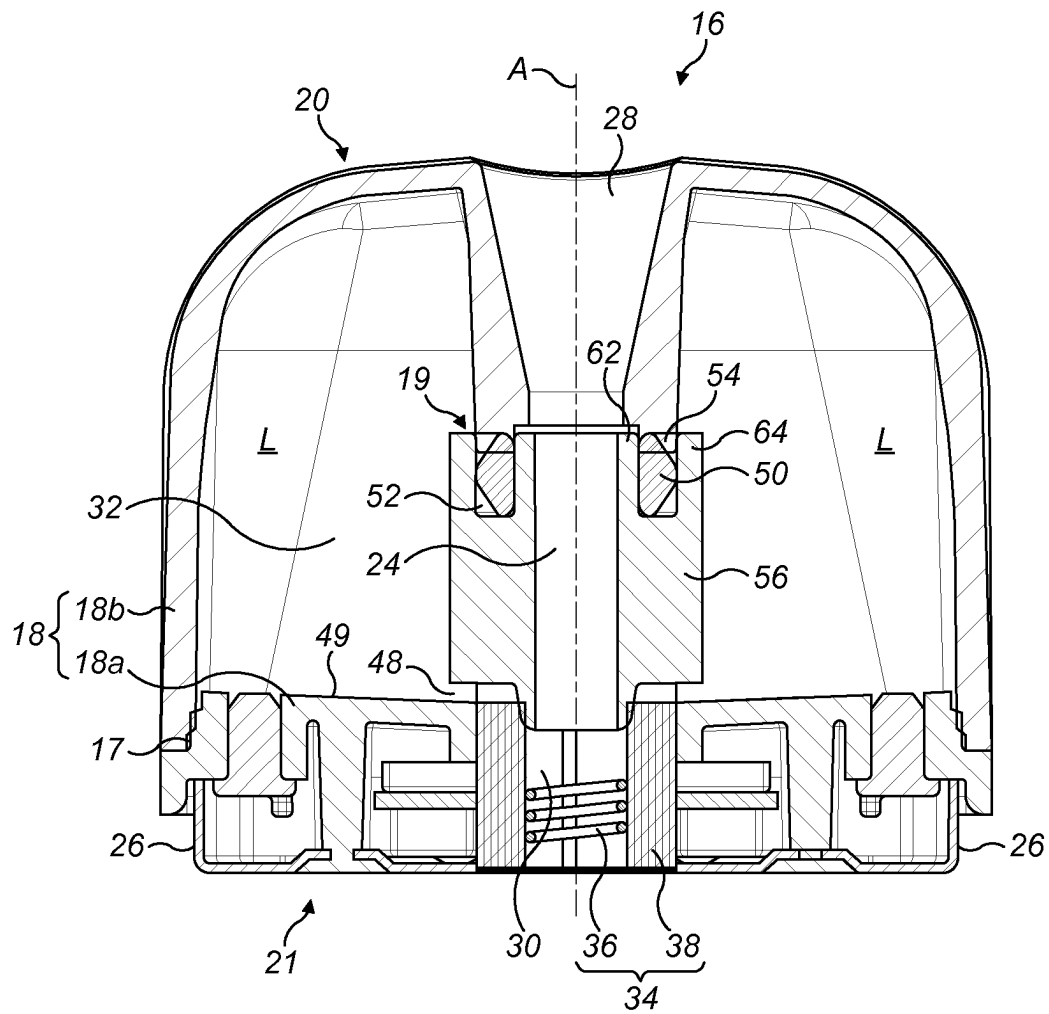


FIG. 5

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**ELECTRONIC CIGARETTE WITH
OPTIMISED VAPORISATION****CROSS-REFERENCE TO RELATED
APPLICATIONS**

The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/EP2019/060542, filed Apr. 24, 2019, published in English, which claims priority to European Application No. EP 18169012.4 filed Apr. 24, 2018, the disclosures of which are incorporated herein by reference.

FIELD OF INVENTION

The present invention relates to personal vaporizing devices, such as electronic cigarettes. In particular, the invention relates to an electronic cigarette and disposable capsules therefor.

BACKGROUND

Electronic cigarettes are an alternative to conventional cigarettes. Instead of generating a combustion smoke, they vaporize a liquid, which can be inhaled by a user. The liquid typically comprises an aerosol-forming substance, such as glycerin or propylene glycol that creates the vapor. Other common substances in the liquid are nicotine and various flavorings.

The electronic cigarette is a hand-held inhaler system, comprising a mouthpiece section, a liquid store, a power supply unit. Vaporization is achieved by a vaporizer or heater unit which typically comprises a heating element in the form of a heating coil and a fluid transfer element. The vaporization occurs when as the heater heats up the liquid in the wick until the liquid is transformed into vapor. The electronic cigarette may comprise a chamber in the mouthpiece section, which is configured to receive disposable consumables in the form of capsules. Capsules comprising the liquid store and the vaporizer are often referred to as "cartomizers".

A problem with electronic cigarettes is that the heater sometimes heats up the liquid such that part of the liquid is transformed to vapor, while another part are brought into a boiling state. This results in that the unvaporized liquid is transformed into larger projections or droplets of liquid that escapes through the mouthpiece. It can be unpleasant for a user to inhale such large droplets, wherefore different ways of alleviating this problem has been proposed.

In order to alleviate this problem, it is common to provide a mesh in the mouthpiece to prevent larger particles from reaching to the user's mouth. The document US20170215481 shows an example of an electronic cigarette having a mesh which avoids larger droplets of liquid to exit through the mouthpiece.

However, as the desired size of the vapor droplets in the aerosol is very small, a mesh still does not give a satisfactory result. Even if the openings in the mesh are reduced, the associated flow restriction would be increased and a satisfactory flow of vapor from the mouthpiece is difficult to achieve.

SUMMARY

In view of the above-mentioned drawbacks of the prior art, it is an object of the present invention to reduce the formation of droplets in the vapor of an electronic cigarette.

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According to a first aspect of the present invention, there is provided a capsule for an electronic cigarette, the capsule having a first end for engaging with an electronic cigarette device and a second end configured as a mouthpiece portion having a vapor outlet, the capsule further comprising: a liquid store configured to contain a liquid to be vaporized, a vaporizing unit comprising a heater and a fluid transfer element, the vaporizing unit being arranged within a vaporizing chamber, a main vapor channel extending from the vaporizing chamber to the vapor outlet in the mouthpiece, and a housing enclosing the liquid store and the vaporizing unit, wherein the fluid transfer element is fluidly connected to the liquid store by at least one liquid inlet and the fluid transfer element provides a capillary action on liquid received therein, wherein the heater is provided at a position that is substantially adjacent the liquid inlet, or at a position between the liquid inlet and the mouthpiece.

Placing the heater at a location which is at a position that is substantially adjacent the liquid inlet or between the liquid inlet and the mouthpiece (and hence generally "above" the liquid inlet when the capsule is in a device and in a "normal" orientation) has the advantage that the amount of liquid around the heater is regulated to an extent by the capillary pressure of the fluid transfer element. In particular, excess quantities of the liquid would tend to form (as a result of a combination of capillary pressure and gravity) within the fluid transfer element below the liquid inlet rather than adjacent thereto or above the liquid inlet.

According to a second aspect of the present invention there is provided a capsule for an electronic cigarette, the capsule having a first end for engaging with an electronic cigarette device and a second end configured as a mouthpiece portion having a vapor outlet, the capsule further comprising: a liquid store configured to contain a liquid to be vaporized, a vaporizing unit comprising a heater and a fluid transfer element, the vaporizing unit being arranged within a vaporizing chamber, a main vapor channel extending from the vaporizing chamber to the vapor outlet in the mouthpiece, and a housing enclosing the liquid store and the vaporizing unit, wherein the housing is composed from an inner housing and an outer housing that are assembled together, wherein the liquid store is located in a void in-between the inner housing and the outer housing, wherein a seal is provided between the inner portion and the outer portion, and wherein the seal has a cross-sectional shape having a cross-sectional height that is larger than a cross-sectional width.

According to a third aspect of the present invention there is provided a capsule for an electronic cigarette, the capsule having a first end for engaging with an electronic cigarette device and a second end configured as a mouthpiece portion having a vapor outlet, the capsule further comprising: a liquid store configured to contain a liquid to be vaporized, a vaporizing unit comprising a heater and a fluid transfer element, the vaporizing unit being arranged within a vaporizing chamber, a main vapor channel extending from the vaporizing chamber to the vapor outlet in the mouthpiece, and a housing enclosing the liquid store and the vaporizing unit, wherein the heater has a height corresponding to 25%-50% of the height of the fluid transfer element, and wherein the convection of the heater is between 4000 and 7000 W/m²K and the power density is between 1.10 to 2.350 Watt/mm², preferably between 1.220 to 2.320 Watt/mm², and more preferably between 1.15 to 1.16 Watt/mm².

Preferably, the fluid transfer element is located within the main vapor channel and has a longitudinal component coinciding with a longitudinal axis of the capsule. In this

way, the capillary action on liquid in the fluid transfer element can be towards the mouthpiece, counteracting the effect of gravity and thereby regulating the flow of liquid from the liquid store to the fluid transfer element. The fluid transfer element can use capillary action to couple liquid away from the liquid inlet. The heater is provided above or adjacent the liquid inlet, and therefore the heater can vaporise liquid that travels within the fluid transfer element using capillary effects. The capillary action can act in the opposite direction to gravity, and this can limit the amount of liquid that is present in the fluid transfer element. This can allow efficient vaporisation of the liquid, and can prevent vaporisation of a saturated fluid transfer element, which may generate unvaporised droplets to the airflow.

Preferably the fluid transfer element is fluidly connected to the liquid store by the at least one liquid inlet, and the external surface of the tubular fluid transfer element abuts the at least one liquid inlet and the internal surface of the tubular fluid transfer element is in contact with the heater.

The liquid inlet may be provided at the bottom of the fluid transfer element, in normal use, at a distance of 0-1 mm from the bottom of the fluid transfer element. The liquid inlets may have a diameter of between 0.8 to 1.3 mm, preferably between 0.95 and 1.15 and more preferably between 1.03 and 1.14 mm. Providing the liquid inlets at the bottom of the fluid transfer element forces the liquid to rise in the fluid transfer element by capillary action. This causes a controlled liquid supply to the heater regardless of the amount of liquid in the liquid store.

The housing preferably comprises an inner housing and an outer housing that are assembled together. The vaporizing chamber is preferably located substantially within the inner portion and the liquid store is preferably located in a void in-between the inner housing and the outer housing. The inner housing and the outer housing may be assembled using a first joint and a second joint, and the second joint may be located radially inwardly of the first joint. The second joint may enable a movement between the inner housing and the outer housing in the axial direction of the capsule such that the relative axial position of the inner housing and the outer housing can be varied.

The inner housing may have a first shoulder and a second shoulder defining a groove there-between. The outer housing may have a protrusion, and the protrusion may be configured to extend into the groove at a variable depth. Preferably the inner housing and the outer housing are sealed together by a compressible seal having a cross-sectional height that is larger than a cross-sectional width. The seal may be provided in the groove defined in the inner housing and it may have a cross-sectional shape that is oval. In other embodiments the seal may have a cross-sectional shape with a transversal projection, projecting in a direction transverse to the axial compressible direction of the seal. The transversal projection may be configured to seal against the inner housing or the outer housing once a compression threshold has been reached.

Preferably the liquid store is configured to maintain a negative pressure such that the flow is regulated and restricted from flowing freely into the fluid transfer element.

The fluid transfer element may have a hollow tubular shape and the heater may be in the form of a heating coil and arranged radially inward of the fluid transfer element. The capillary height of the fluid transfer element preferably exceeds the axial height of the heating coil. In some embodiments the heating coil has a height corresponding to 25%-50% of the height of the fluid transfer element, preferably 25%-45% or most preferably 35%. The fluid transfer ele-

ment may have a capillary height corresponding to the actual height of the fluid transfer element. The height of the fluid transfer element may be between 4.5 and 6.5 mm and the height of the heating coil may be 1.8 to 2.5 mm, preferably 5.8 mm and 2.04 mm respectively.

Preferably the convection of the heater is between 4000 and 7000 W/m²K, preferably between 5500 and 6500 W/m²K, and most preferably between 5800 W/m²K and 6200 W/m²K. In this way, it has been found that, due to the latent heat of vaporisation, the energy produced by the heater causes vaporisation in the fluid transfer element and drives the vapour off, rather than raising the temperature of the liquid in the liquid store.

The heater may be a heating coil with a number of turns between 2 to 4, preferably 3 turns. In some embodiments the heating coil may be titanium.

The present invention is based on a realization of the inventors that droplets in the vapor can be reduced by improving the vaporization capabilities of an electronic cigarette. The projections of liquid droplets are often caused when the liquid enters a boiling state instead of a vaporization state. By reducing the boiling effect in the vaporizing chamber and increasing the vaporization capabilities, more liquid can be brought into the vaporization stage.

Each aspect of the invention has the desirable property of reducing the formation of liquid projections. However, if the solutions are used in combination, the effects from the functional group of features is added to each other and synergies can be achieved. Therefore, features of one aspect of the invention can be combined with any other aspect of the invention.

According to an embodiment, there is provided a capsule for an electronic cigarette, the capsule having a first end for engaging with an electronic cigarette device and a second end configured as a mouthpiece portion having a vapor outlet, the capsule further comprising: a liquid store configured to contain a liquid to be vaporized, a vaporizing unit comprising a heater and a fluid transfer element, the vaporizing unit being arranged within a vaporizing chamber, an air inlet or inlets, a main vapor channel in fluid communication with the air inlet or inlets at one end and with the vapor outlet in the mouthpiece at the other end and incorporating the vaporizing chamber, and a housing enclosing the liquid store and the vaporizing unit, wherein the fluid transfer element is fluidly connected to the liquid store by at least one liquid inlet and the fluid transfer element provides a capillary action on liquid received therein, wherein the fluid transfer element extends in a direction along the main vapor channel in one or both directions away from the liquid inlets by an amount which exceeds the extension of the heater along the main vapour channel.

Preferably, the fluid transfer element is configured as a tube, the external surface of which abuts the at least one liquid inlet, and the internal surface of which is in contact with the heater. Preferably the heater is located within the fluid transfer element adjacent to the at least one liquid inlet. In this way, when the fluid transfer element adjacent to the heater becomes dry as a result of liquid to be vaporized being vaporized by the heater, liquid flows by capillary action both radially through the fluid transfer element from the or each liquid inlet and additionally by capillary action (preferably with gravitational assistance if the device is held in a normal usage orientation in some embodiments) axially and/or circumferentially through the fluid transfer element from other portions of the fluid transfer element, thus enabling quick and efficient replenishment of liquid to be vaporized to portions of the fluid transfer element in contact

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with the heater. This prevents portions of the heater from becoming dry as a result of insufficient replenishment of liquid to portions of the fluid transfer element in contact with the heater element and remote from sources of re-supply of liquid to be vaporized (whether those sources of resupply are (other parts of) the fluid transfer element or the liquid inlet or inlets) without requiring very numerous or large (in terms of surface area) inlets—this is advantageous because using numerous or large inlets can cause problems with leakage of liquid through the fluid transfer element, especially where the fluid transfer element dries out adjacent a liquid inlet (or a portion of a liquid inlet) on one side of the liquid inlet, and the surface of the liquid in the liquid reservoir also falls below the liquid inlet (or portion thereof) on the other side of the liquid inlet, as this can permit air at atmospheric pressure to leak into the liquid reservoir, through the “dry” fluid transfer element and into the space above the surface of the liquid, thus destroying the negative pressure in the space above the liquid surface in the liquid reservoir which can lead to (increased) undesirable leakage of liquid through the fluid transfer element and into the vaporization chamber.

In some embodiments, the heater is provided at a position that is substantially adjacent the liquid inlet. This is advantageous as it minimizes the distance that liquid needs to travel to get from the inlet(s) to the heater. As a result, liquid can travel along different resupply routes to travel to portions of the fluid transfer element in contact with the heater to maximize the efficiency of the liquid resupply. This is in contrast to conventional arrangements in which resupply routes through the fluid transfer element often merge resulting in slower resupply. It will be appreciated that resupply of liquid from other parts of the fluid transfer element which are more remote from the liquid inlet(s) than the heater, will not (contrary to prior art arrangements) tend to be resupplied themselves with liquid from the liquid store until after the portions of the wick adjacent the heater have themselves been resupplied. This arrangement works well in e-cigarettes as there is usually ample time between puffs for liquid to be resupplied to the whole of the fluid transfer element. Thus these remoter areas can act as buffers enabling quick resupply to certain parts of the wick during a puff, the buffers then being refilled in-between puffs.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will now be described with reference to the appended drawings, which by way of example illustrate embodiments of the present invention and in which like features are denoted with the same reference numerals, and wherein:

FIG. 1a is a schematic perspective view of an inhaler and a capsule according to an exemplary embodiment of the present invention;

FIG. 1b is a schematic perspective view of the inhaler and capsule of FIG. 1a and in which the front panel of the inhaler has been removed;

FIG. 1c is a schematic perspective view of the inhaler in FIGS. 1a and 1b, wherein the back panel of the inhaler has been removed;

FIG. 2a is a schematic front cross-sectional view of a capsule according to an embodiment of the present invention;

FIG. 2b is a schematic side cross-sectional side-view of a capsule according to an embodiment of the present invention;

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FIG. 2c is a schematic side cross-sectional side-view of a capsule according to another embodiment of the present invention;

FIGS. 3a to 3d are cross-sectional views of capsule seals according to embodiments of the present invention;

FIG. 4a is a schematic exploded view of a capsule of the present invention;

FIG. 4b is a schematic cross-sectional view of the inner housing of the capsule of FIG. 3c; and

FIG. 5 is a cross-sectional view of a capsule in an embodiment of the invention.

DETAILED DESCRIPTION

As used herein, the term “inhaler” or “electronic cigarette” may include an electronic cigarette configured to deliver an aerosol to a user, including an aerosol for smoking. An aerosol for smoking may refer to an aerosol with particle sizes of 0.5-7 microns. The particle size may be less than 10 or 7 microns. The electronic cigarette may be portable.

Referring to the drawings and in particular to FIGS. 1a to 1c, an electronic cigarette 2 for vaporizing a liquid L is illustrated. The electronic cigarette 2 can be used as a substitute for a conventional cigarette. The electronic cigarette 2 has a main body 4 comprising a power supply unit 6, electrical circuitry 8 and a capsule seating 12. The capsule seating 12 is configured to receive removable capsules 16 comprising a vaporizing liquid L.

The capsule seating 12 is in the form of a cavity configured to receive the capsule 16. The capsule seating 12 is provided with a connection portion 21 configured to hold the capsule 16 firmly to the capsule seating 12. The connection portion 21 could for instance be an interference fit, a snap fit, a screw fit, a bayoneted fit or a magnetic fit. The capsule seating 12 further comprises a pair of electrical connectors 14 configured to engage with corresponding power terminals 45 on the capsule 16.

As best seen in FIGS. 2a and 2b, the capsule 16 comprises a housing 18, a liquid store 32, a vaporizing unit 34 and power terminals 45. The housing 18 has a mouthpiece portion 20 provided with a vapor outlet 28. The mouthpiece portion 20 may have a tip-shaped form to correspond to the ergonomics of the user's mouth. On the opposite side of mouthpiece portion 20, the connection portion 21 is located. The connection portion 21 is configured to connect with the connector in the capsule seating 12. In the illustrated embodiment of FIGS. 2a and 2b, the connection portion 21 on the capsule 16 is a metallic plate, configured to connect to a magnetic surface in the capsule seating 12. The capsule housing 18 may be in a transparent material, whereby the liquid level of the capsule 16 is clearly visible to the user. The housing 18 may be formed in a polymeric or plastic material, such as polyester.

The vaporizing unit 34 comprises a heating element 36 and a fluid transfer element 38. The fluid transfer element 38 is configured to transfer the liquid L by capillary action from the liquid store 32 to the heating element 36. The fluid transfer element 38 can be a fibrous or porous element such as a wick made from twined cotton or silica. Alternatively, the fluid transfer element 38 can be any other suitable porous element.

A vaporizing chamber 30 is defined in the area in which liquid vaporization occurs and corresponds to the proximal area in which the heating element 36 and the fluid transfer element 38 are in contact with each other. The fluid transfer element 36 has an upper distal end 38a and a lower distal end

38b. The lower distal **38b** end is provided at the lower end of the vaporizing chamber **30**. The vaporizing chamber **30** is located at the opposite distal end of the capsule **16** to the mouthpiece portion **20**. From the vaporizing chamber **30** to the vapor outlet **28** in the mouthpiece portion **20**, a main vapor channel **24** is formed and may have a tubular cross-section. The main vapor channel **24** is thus extending from the vaporizing chamber **30** to the vapor outlet **28** in the mouthpiece portion **20**. The vaporizing chamber **30** has a bottom surface **46** arranged opposite of the vapor outlet **28**. The bottom surface is a liquid impermeable surface, which closes the vaporization chamber **30**.

The liquid **L** may comprise an aerosol-forming substance such as propylene glycol or glycerol and may contain other substances such as nicotine. The liquid **L** may also comprise flavorings such as e.g. tobacco, menthol or fruit flavor.

As seen in FIGS. **4a** and **4b**, the vaporizing chamber **30** is fluidly connected to the liquid store **32** using at least one liquid inlet **48**. The liquid inlet **48** is arranged at the bottom surface **46** of the liquid store **32**, at a distance of 0-2 mm above the bottom surface **46**, preferably 0-1 mm. The position of the liquid inlets **48** close to the bottom surface **46** of the liquid store **32** avoids liquid **L** from the liquid store **32** from flowing freely into the vaporization chamber **30**. The liquid inlet **48** is also located close to the lower distal end **38b** of the fluid transfer element **38**. The liquid inlets **48** are thus located 1-3 mm from the lower distal end **38b** of the fluid transfer element **38**, preferably 1-2 mm. The heating element **36** is advantageously positioned with its first contact approximately aligned with the liquid opening, that is in line with or 1 mm below the liquid inlets or 1-2 mm above the liquid inlets. Preferably, the heating element **36** is in contact with the fluid transfer element **38**. If the liquid **L** flows freely, there is a risk of oversaturating the fluid transfer element **38**. The liquid inlets close to the bottom surface **46** of the liquid store **32** enables a negative pressure to form in the liquid store **32** during vaporization and until the liquid store **32** gets empty. This is because the liquid inlets **48** are positioned vertically underneath the liquid surface **S** in the capsule **16** until the capsule **16** is close to depletion. The close to depletion can be defined as when the volume of liquid **L** in the capsule **16** has decreased with 90% from the original volume. This is achieved when the electronic cigarette **2** is in an essentially upright position and thus during normal usage of the electronic cigarette **2**.

As illustrated in FIGS. **1a** and **1b**, the capsule **16** may have a shape that is not rotationally symmetrical in the axial direction. The capsule **16** may therefore have a rectangular base with flat longer side and a short side. This shape may also correspond to the shape of the electronic cigarette **2**. The liquid inlets **48** may advantageously be provided in the short side of the capsule **16**. This maintains a negative pressure in the liquid store **32** as the liquid inlets **48** remain below the surface of the liquid surface when the electronic cigarette is in a resting position (lying flat on a surface such as a table). This effect lasts at least until the liquid store **32** is about half-full. Additionally, even when the liquid store **32** is less than half-full, while the fluid transfer element is "wet" it effectively seals against air passing through the fluid transfer element and reducing the negative pressure. Typically, because of gravity, "drying" of the fluid transfer element or wick will start at the top of the wick and only slowly migrate downwards. Therefore even when the liquid store is less than half-full placing the liquid inlets so as to not be located at the top of the fluid transfer element when the electronic cigarette is in a resting position, still assists in maintaining the negative pressure.

The bottom surface **49** of the liquid store **32** may also be provided with a downwardly sloping surface **49** against the at least one liquid inlet **48**. The downwardly sloping surface **49** enables all liquid **L** in the liquid store **32** to be transported towards the liquid inlet **48** and to be further absorbed by the fluid transfer element **38** inside the main channel **24**. The capsule **16** is further provided with at least one air intake channel **26** extending from a first opening in the capsule **16**, to the vaporizing chamber **30**.

As best seen in FIGS. **2a**, **2c** and **4a**, the capsule housing **18** may be formed from an inner housing **18a** and an outer housing **18b** assembled together with the liquid store **32** located in a void in-between the inner housing **18a** and the outer housing **18b**. The inner housing **18a** and the outer housing **18b** may be assembled using a first joint **17** and a second joint **19**. The first joint **17** is located at the bottom portion of the capsule **16** and may advantageously be achieved by ultrasonic welding.

The second joint **19** is located inside the capsule **16** and can be achieved by a seal **50** housed inside a circular groove **52** in the inner housing **18a**. The inner housing **18a** has a first shoulder **62** and a second shoulder **64** defining the groove **52** there-between. The outer housing **18b** is provided with a projection **54**, which is configured to extend into the groove **52** at a variable depth. The projection **54** is arranged to abut against the seal **50**. As the seal **50** is compressible in the axial direction **A** of the capsule **16**, the projection **54** may enter the groove **52** at a variable depth.

The inner housing **18a** is configured to house the vaporizing unit **34**, which is located in the main channel **24** extending from the bottom surface **46** of the vaporization chamber **30**, as previously described. In order to avoid that the fluid transfer element **38** collapses into the vaporization chamber **30**, the inner housing **18a** may be provided with a flange **56**, which is encircling the inner circumference of the fluid transfer element **38**.

The inner housing **18a** comprises a tubular column or chimney **80** extending from the at least one fluid inlet **48** to the first shoulder **62**. The tubular column **80** is provided radially outwardly of the fluid transfer element **38** so that it provides structural support to the fluid transfer element **38**. The flange **56** that encircles the inner circumference of the fluid transfer element **38** is attached the tubular column by a radial strut **82**. In this way, the tubular column **80** can provide structural support to the internal and external surfaces of the tubular fluid transfer element **38**.

As may be appreciated from FIG. **2a** in particular, the first shoulder **62** is provided as part of the tubular column **80**. The second shoulder **64** is connected to the tubular column **80** by the radial strut **82** so that the annular groove **52** is defined between the first and second shoulders **62**, **64**.

An advantage of having the two-part housing **18** comprising the inner housing **18a** and the outer housing **18b** is that the assembly of the internal parts of the vaporization unit **34** is facilitated. However, as the capsule **16** is assembled by a first housing **18a** and the second housing **18b**, there may be variations in the manufacturing process. The seal **50** is therefore configured to accommodate for variations in the manufacturing process.

Because the inner housing **18a** and the outer housing **18b** are sealed together, a negative pressure forms in the liquid store **32** when fluid flows out of the liquid store **32**. The negative pressure regulates the liquid flow from the liquid store **32** to the fluid transfer element **38**. The negative pressure thus creates a resistance to free flow of liquid **L** into the vaporization chamber **30** and in that way regulates the liquid flow. The at least one fluid inlet **48** can be provided at

the end portion of the heating element 36 in its most proximal point to the base of the capsule 16.

FIG. 3a illustrates a conventional O-ring with a circular cross section. The seal 50 of FIG. 3a can be used in the capsule 16 according to the present invention. However, as seen in FIGS. 3b, 3c and 3d, the seal 50 may have a cross-sectional height h_s that is larger than the cross-sectional width w_c . This provides an advantage of that the seal 50 is configured to accommodate for a longer axial variations between the position of the inner housing 18a in relation to the outer housing 18b, while maintaining a compact shape in the transverse direction.

In the embodiment illustrated in FIGS. 3b, 3c and 3d, the seal 50 is provided with a non-circular shape, such that the seal is longer in the axial direction (coinciding with the axial direction of the capsule 16). The seal 50 can have a rectangular cross-section as illustrated in FIGS. 2c and 3d.

In the embodiment illustrated in FIG. 3b, in which the seal 50 has a T-shaped form. The T-shape provides the same advantage in terms of the long accommodation for axial differences. As an additional effect, the transversal protrusion 58 enables the seal 50 to additionally seal against the first shoulder 62 and the second shoulder 64.

The long cross-sectional height h_s of the oval and t-shaped seals 50 provides for a long deformation length and a long distance throughout which the seal 50 is capable of sealing the inner housing 18a and the outer housing 18b against each other. Additionally, the relatively small width of the seal 50 reduces the space of the seal 50 in the horizontal direction such that the size of the capsule 16 and the liquid content L in the liquid store can be optimized.

The O-ring with a circular cross-section provides a sealing effect between the inner housing 18a and the outer housing 18b. Because of the variations in the ultrasonic welding process, the seal is configured to accommodate a difference of ± 0.5 mm. The oval seal and T-shaped seals provide a longer compression distance through which a sealing effect is achieved.

The circular, the oval, the rectangular and the T-shaped seals demonstrate different compression behavior, i.e. the seals present different resistance to an axial deformation force F_c . This behavior is related to the geometric differences in the horizontal cross-sectional area and the vertical height of the seals. Hence, the geometric differences translate into different spring constants among the circular, oval and T-shaped seals. The spring constant for the seals also varies in a non-linear manner as the cross-section of the seals present different cross-sectional areas in the axial direction thereof. When the compression force F_c divided by the cross-sectional area, the force distributes over the cross-sectional area and can be measured in Newton/m².

For the oval in comparison with the seal circular seal, the cross sectional area is smaller in relation to the vertical height. This means that the oval seal has a lower elasticity module than the circular seal and thus acts much more flexible.

The T-shaped seal also has a similar cross-sectional area as the oval seal. However, the T-shaped seal provides for a first region of high compressibility (low spring constant) and a second region over the horizontal T-shaped protrusion with stiffer region (of a higher spring constant). The T-shaped protrusion provides another benefit, which is to in addition seal against a lateral surface.

Now referring to FIGS. 2a and 2b in which it is illustrated that the fluid transfer element 38 may have a tubular form and have an axial longitudinal direction coinciding with the axial longitudinal direction of the main channel 24. The

tubular form provides a vapor channel 40 inside the fluid transfer element 38, through which the vapor can leave the vaporizing chamber 30 to travel to the vapor outlet portion 28. Furthermore, the tubular form of the fluid transfer element 38 also provides a snug fit against the inner wall of the main channel 24 and forms a space therein for receiving the heating element 36.

The heating element 36 may advantageously be in the form of a coil-shaped heater 36 and be aligned with its axial direction coinciding with the longitudinal direction of the fluid transfer element 38. Hence, a coil-shaped heater 36 can be fitted into the vapor channel 40 defined inside the fluid transfer element 38 while providing a close contact with the fluid transfer element 38. In such a way, the fluid transfer element 38 can be retained in-between the inner wall of the main channel 24 and the heating element 36. This also helps the fluid transfer element 38 to maintain its shape and avoid collapsing. The material of the fluid transfer element 38 can be cotton, silica, or any other fibrous or porous material.

The heating element 36 is provided with a height corresponding to a proportion of the capillary height of the fluid transfer element 38. The inventors have found that if the heating element 36 is provided with a height largely exceeding the capillary height of the fluid transfer element 38, the heating element 36 tends to be in contact with a dry top portion of the fluid transfer element 38 as the liquid level in the liquid store 32 becomes depleted. The fluid transfer element 38 in the bottom portion of the capsule 16 is often saturated or even over-saturated with liquid while the upper portion of the fluid transfer element 38 is left dry. If heat is applied to the fluid transfer element 38, the temperature of the heating element 36 at the dry portion of the fluid transfer element 38 is not cooled off by the surrounding liquid L, whereby the dry portion is excessively heated. In the over-saturated portion of the fluid transfer element 38, the temperature is lower and boiling bubbles and projections can be formed. The heat from the vaporizing unit 34 is transferred inside the liquid store 32 and parts of the capsule 16. It is therefore advantageous to avoid formation of local variations and presence of dry areas of the fluid transfer element 38 in contact with the heating element 36.

On the other hand, if the capillary height of the fluid transfer element 38 largely exceeds the height of the heating element 36, the heating element 36 will become oversaturated along its entire axial length and the temperature of the heating element 36 is cooled down rather than achieving an efficient vaporizing the liquid. This may again lead to bubble formation and liquid projections, while the temperature increases in the liquid storage portion 32 and the housing of mouthpiece portion 20. In the typical vaporization process of an electronic cigarette, the vaporization is achieved by boiling of the liquid below the surface of the liquid. If the level of saturation of the heating element 36 is kept at an ideal level, such that the heating element 36 is only covered with a small amount of liquid L, the boiling does not create large projections of liquid, but instead creates a uniform heating of the liquid and enables the liquid to go directly into a vapor state.

It is common to detect the temperature of the heating element 36, as the temperature of the heating element 36 increases when the fluid transfer element 38 gets dry. In the absence of fluid around the heating element 36, the temperature of the heating element 36 increases. This is because fluid present around the heating element 36 absorbs energy from the heating element 36 when it passes into a vaporization state, which results in a cooling effect on the heating element 36. That is to say, heat from the heating element 36

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tends to be used to provide the latent heat of vaporization required to transform the liquid into gas at the boiling point temperature, rather than causing the temperature of the heating element **36** and any surrounding material to increase in temperature. By measuring the temperature of the heating element **36**, the vaporization temperature can be controlled so that the fluid transfer element **38** is not overly heated.

An ideal vaporization is characterized by a high vapor volume, a minimal amount of heat transferred to the liquid store and a low presence of liquid projections.

A first exemplary prototype was designed based on previously known configurations and relative dimensions of a heater element **36** and fluid transfer element **38** combination. In a first example, the following parameters were selected:

Example 1

Diameter: 0.4 mm
Resistive length: 70 mm
Resistance: 0.294 Ω
Total effective length: 68 mm
Pitch: 0.7 mm
Heating coil height: 4.75 mm
Total effective surface: 85.45 mm²
Power density: 0.187 W/mm²
Convection heated up: 1040 W/m²K
Height of fluid transfer element: 5.8 mm

Additionally, liquid inlets to from the fluid transfer element **38** were spread out in the axial direction of the fluid transfer element **38** in order to provide a sufficient liquid supply along the entire length of the heater element **36**.

However, the first exemplary capsule provided an unsatisfactory result, despite the sufficient and well distributed liquid supply to the heater element **36** and saturated fluid transfer element **38**. The coil presented an inconsistent heating profile, where the lower part of the heating coil reached only up to 300 K and where the upper part of the coil reached up to about 900 K. As the total measurable resistance corresponds to the sum of the resistance over the whole coil length, the temperature could not be regulated on the basis of a resistance measurement, as the temperature was not consistent over the entire coil length.

With a background of the problems of the first example of the coil, the inventors have found that the lower section of the fluid transfer element **38** could be configured with a wetted height h_w , as long as there is liquid left in the liquid store **32**. The wetted height corresponds to the distance capillary action will take place. The heating element **36** should therefore be relatively short in order to not extend above the upper (dry) section of the fluid transfer element **38**. However, the heating element **36** still needs to be configured to produce a satisfying amount of vapor. The fluid transfer element **38** should be supplied with a controlled and consistent amount of liquid. Hence, the liquid supply rate needed to be controlled during the vaporization. The liquid inlets were the bottom of the fluid transfer element **38** forces the liquid to rise in the fluid transfer element **38** by capillary action. This causes a controlled liquid supply to the heater element **36** regardless of the amount of liquid in the liquid store.

Moreover, advantageous dimensions found by the inventors include a height of the fluid transfer element **38** of between 4.5 and 6.5 mm and a height of the heating coil of between 1.8 to 2.5 mm. Preferably the height is 5.8 mm and 2.04 mm respectively.

Preferably, the height of the heating coil **36** in relation to the fluid transfer element is 20-50%, preferably between

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25% and 45% and most preferably around 35% of the height of the fluid transfer element **38**. The porous material of the fluid transfer element **38** is preferably selected such that the capillary height of the fluid transfer element is equal to the actual height of the fluid transfer element. The capillary height of the fluid transfer element **38** can even exceed the actual height of the fluid transfer element. In this case, we can refer to a theoretical capillary height.

Compared to the first example initial and standard configuration of a heating coil **36** and fluid transfer element **38** configuration, the height of the heating element **36** was reduced to approximately a half of the initial height. The height was reduced to various levels in the different samples. In absolute measures, the height of the heating element (i.e. the heating coil **36**) was reduced with at least 3 mm. An advantage having a long wick is that it can retain a reserve of liquid and thus act as a buffer. The wick can is therefore adapted to supply liquid to wick in the heater region, if for instance the electronic cigarette is held upside down. Additionally, as discussed above, the buffer also provides an independent resupply route through the fluid transfer element **38** for resupplying liquid to the portions of the fluid transfer element **38** during a puff even when the electronic cigarette **2** is held in a normal orientation.

The inventors found that the liquid flow from the liquid store **32** needs to be precisely matched to the power density in order to get a high level of vapor production, avoid dry fluid transfer element **38**, formation of bubbles and excessive heating of the liquid in the liquid store **32**. It was a surprising effect that by increasing the power density, the liquid temperature in the liquid store **32** was found to decrease. During the tests, it was found that by increasing the convection from 1900 W/m²K to 6000 W/m²K, and the power density from 0.187 W/mm² to 1.152 W/mm², the temperature in the liquid store was reduced from 108° C. to 54° C. The increased convection and power density were achieved by increasing the resistance of the heating coil by reducing its diameter.

In order to verify the interrelationship of the fluid transfer rate and the power density, a number of capsule prototypes were tested. The target convection of the heating element **36** was found to be between 5000 and 7000 W/m²K, preferably between 5500 W/m²K and 6500 W/m²K and most preferably at 6000 W/m²K.

When the height of the heating element **36** was reduced, the diameter of the heating coil was also reduced in order to obtain the desired convection of 6000 W/m²K. Hence, the height was decreased to further increase the power density of the heating coil for the same amount of power applied to the heating coil. However, it was shown that the heating wire forming the heating coil **36** cannot be permitted to become too thin for two principal reasons: firstly, the coil **36** can become mechanically weak which makes it difficult to assemble and it ceases being able to support the fluid transfer element **38** and prevent its deformation into the main vapor channel **40**. This is undesirable as the vapor channel diameter is an important parameter affecting device performance and it is therefore important to have consistent control over this parameter which is difficult to achieve if the fluid transfer element **38** is partially blocking the vapor channel, and secondly, as the heating wire becomes thinner the effect of manufacturing tolerances in the wire thickness have a greater impact and some portions of the wire can become very thin—these portions are then at risk of overheating relative to other portions of the wire and possibly fusing.

In order to reduce the height and still achieve the same power density, the coil diameter was nonetheless reduced

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and different values were assessed. The optimum coil diameter was then selected from among the values 0.4, 0.3, 0.254 and 0.226 mm.

The result of the assessment was that an optimized capsule as per Example 2, which could have:

Example 2

Diameter: between 0.226 and 0.3 mm, preferably 0.254 mm

Resistive length: 26.92 mm

Resistance: between 0.291 to 0.295 Ω

Total effective length: 26.09 mm

Pitch: between 0.5-1.0 mm, preferably 1.0 mm

Heating coil height: between 2.4-3.2 mm

Total effective surface: 20.82 mm²

Power density: between 1.152 to 2.319 Watt/mm², preferably 1.152 Watt/mm²

Convection heated up: between 5000 and 7000 W/m²K, preferably around 6000 W/m²K, W/m²K

Height of fluid transfer element: between 4.5 and 6.5 mm, preferably 5.8 mm mm

Capillary height of the fluid transfer element: same or exceeding the actual height of the fluid transfer element

Sealing type: Shown that a non-circular seal with larger height than width was the most advantageous to maintain a negative pressure in the liquid store 32.

The optimum pitch of the windings was found to be with in a preferred range of between 0.5-1.0 mm to ensure a satisfactory heat distribution.

The target heating temperature for the second exemplary capsule was the same as for the first exemplary capsules, which was 270° C.

It was also found that the number of windings of the heating coil 36 should preferably be between 2 and 4, and most preferably 3. Having a number of windings between 2 and 4 provide a heating coil 36 that is less flimsy and can better hold together in the manufacturing process of the heating coil 36. Additionally, having three coil windings is very efficient in terms of the resupply routes of the liquid to the portions of the fluid transfer element 38 in contact with the heating element 36. In particular, there is a direct path radially through the liquid inlets 48 towards the centre coil of the heater. Additionally, some liquid from the liquid inlets 48 can travel downwards towards the bottom coil winding of the heating element 36. Simultaneously a minor resupply route is provided from the portion of the fluid transfer element 38 immediately below the bottom coil. A major resupply route is from the portion of the fluid transfer element above the top coil to the portion of the fluid transfer element in contact with the top coil of the heating element 36. Only a small amount of liquid from the liquid inlets will travel upwards to resupply this portion as it is mostly resupplying the liquid vaporized by the middle and lower coil windings, so most of the resupply liquid comes from the buffer portion above the top coil winding. This is then resupplied by capillary action in-between puffs.

An advantage of having a fluid transfer element 38 having a height greater than the heating element 36 and also having a correspondingly high capillary height is that the size of the liquid inlets 48 can be minimized as the liquid inlets can be configured such that they only need to resupply a portion of the liquid being vaporized during a puff as the liquid in the fluid transfer element 38 can supplement liquid passing through the liquid inlet(s) 48 for resupplying vaporized liquid during a puff. Naturally, the size of the liquid inlets needs to be determined in view of the viscosity of the liquid

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to be used in the liquid store 32. The dimensions in this embodiment are chosen to be optimal for use with liquid comprising mostly a mixture of Vegetable Glycerin (VG) and Propylene Glycol (PG) with ratios of between 40 and 60% (i.e. ranging from VG:PG=40:60 to VG:PG=60:40). The dimensions of the inlets would naturally be increased slightly if using higher proportions of VG (e.g. up to substantially 100% VG and no PG) due to the greater viscosity of VG compared to PG.

FIG. 5 is a cross-sectional view of a capsule 16 in another embodiment of the invention. The capsule 16 differs from the arrangement shown in FIG. 2A in the position of the vaporisation chamber 30. In this arrangement the vaporisation chamber 30 is positioned entirely below the liquid store 32. Liquid inlets 48 are provided in the base of the liquid store 32, fluidly connecting the liquid store 32 with the fluid transfer element 36. The capillary action in the fluid transfer element 36, together with the downward force of gravity, can encourage liquid in the liquid store 32 to flow into the fluid transfer element 36. The flow of liquid is regulated in this arrangement by a negative pressure that forms in the liquid store 32 when the liquid is drained. The heating coil 36 includes three coils in this arrangement, and it is provided radially inwardly of the fluid transfer element 38.

The skilled person will realize that the present invention by no means is limited to the described exemplary embodiments. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage. Moreover, the expression “comprising” does not exclude other elements or steps. Other non-limiting expressions include that “a” or “an” does not exclude a plurality and that a single unit may fulfill the functions of several means. Any reference signs in the claims should not be construed as limiting the scope. Finally, while the invention has been illustrated in detail in the drawings and in the foregoing description, such illustration and description is considered illustrative or exemplary and not restrictive; the invention is not limited to the disclosed embodiments.

The invention claimed is:

1. A capsule for an electronic cigarette, the capsule having a first end for engaging with an electronic cigarette device and a second end configured as a mouthpiece portion having a vapor outlet, the capsule further comprising:

- a liquid store configured to contain a liquid to be vaporized,
- a vaporizing unit comprising a heater and a tubular fluid transfer element, the vaporizing unit being arranged within a vaporizing chamber,
- a main vapor channel extending from the vaporizing chamber to the vapor outlet in the mouthpiece portion, and
- a housing enclosing the liquid store and the vaporizing unit,

wherein the tubular fluid transfer element is fluidly connected to the liquid store by at least one liquid inlet which abuts an external surface of the tubular fluid transfer element and the tubular fluid transfer element provides a capillary action on liquid received therein, wherein the heater is provided in contact with an internal surface of the tubular fluid transfer element and at a position between the at least one liquid inlet and the mouthpiece portion, wherein the housing comprises an inner housing and an outer housing that are assembled together, wherein the vaporizing chamber is located substantially within the inner housing and the liquid store is located in a void in-between the inner housing

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and the outer housing, and wherein the inner housing is provided with a flange that encircles an inner circumference of the tubular fluid transfer element to prevent the tubular fluid transfer element collapsing into the vaporizing chamber,

wherein the inner housing and the outer housing are assembled using a first joint and a second joint, wherein the second joint is located radially inwardly of the first joint, and wherein the second joint enables a movement between the inner housing and the outer housing in an axial direction of the capsule such that a relative axial position of the inner housing and the outer housing is configured to be varied,

wherein the inner housing has a first shoulder and a second shoulder defining a groove there-between, wherein the outer housing has a protrusion, and wherein the protrusion is configured to extend into the groove at a variable depth along the axial direction

wherein the inner housing and the outer housing are sealed together by a compressible seal having a cross-sectional height that is larger than a cross-sectional width thereof, and

wherein the seal has a cross-sectional shape with a transversal projection, projecting in a direction transverse to an axial compressible direction of the seal, wherein the transversal projection is configured to seal against the inner housing or the outer housing once a compression threshold has been reached and wherein the cross-sectional shape of the seal is asymmetrical about any axis parallel to the axial compressible direction of the seal, and

wherein the compressible seal is entirely contained within the groove.

2. The capsule according to claim 1, wherein the cross-sectional shape of the seal has a top extremity and a bottom extremity such that the cross-sectional shape has a longitudinal axis extending between the top and bottom extremities, and wherein the projection projects further from the longitudinal axis in the direction transverse to the axial compressible direction than any other portion of the cross-sectional shape projects from the longitudinal axis.

3. A capsule for an electronic cigarette, the capsule having a first end for engaging with an electronic cigarette device and a second end configured as a mouthpiece portion having a vapor outlet, the capsule further comprising:

- a liquid store configured to contain a liquid to be vaporized,
- a vaporizing unit comprising a heater and a tubular fluid transfer element, the vaporizing unit being arranged within a vaporizing chamber,
- a main vapor channel extending from the vaporizing chamber to the vapor outlet in the mouthpiece portion, and
- a housing enclosing the liquid store and the vaporizing unit,

wherein the tubular fluid transfer element is fluidly connected to the liquid store by at least one liquid inlet which abuts an external surface of the tubular fluid transfer element and the tubular fluid transfer element provides a capillary action on liquid received therein, wherein the heater is provided in contact with an internal surface of the tubular fluid transfer element and at a position that is substantially adjacent to the at least one liquid inlet, or at a position between the at least one liquid inlet and the mouthpiece portion, wherein the housing comprises an inner housing and an outer housing that are assembled together, wherein the vaporizing chamber is located substantially within the

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inner housing and the liquid store is located in a void in-between the inner housing and the outer housing, and wherein the inner housing is provided with a flange that encircles an inner circumference of the tubular fluid transfer element about an axis to prevent the tubular fluid transfer element from collapsing into the vaporizing chamber

wherein the inner housing has a first shoulder and a second shoulder defining a groove there-between, wherein the outer housing has a protrusion, and wherein the protrusion is configured to extend into the groove at a variable depth

wherein the inner housing and the outer housing are sealed together by a compressible seal having a cross-sectional shape that is irregular with a transversal projection which projects transverse to the axis, and wherein the compressible seal is entirely contained within the groove.

4. The capsule according to claim 3, wherein the tubular fluid transfer element is located within the main vapor channel and has a longitudinal component coinciding with a longitudinal axis of the capsule, whereby the capillary action on liquid in the tubular fluid transfer element is towards the mouthpiece portion, counter-acting an effect of gravity and thereby regulating a flow of liquid from the liquid store to the tubular fluid transfer element.

5. The capsule according to claim 3, wherein the at least one liquid inlet is provided at a bottom of the tubular fluid transfer element, at a distance of 0-1 mm from the bottom of the tubular fluid transfer element.

6. The capsule according to claim 3, wherein the at least one liquid inlet has a diameter of between 0.8 to 1.3 mm.

7. The capsule according to claim 3, wherein the inner housing and the outer housing are assembled using a first joint and a second joint, wherein the second joint is located radially inwardly of the first joint, and wherein the second joint enables a movement between the inner housing and the outer housing in an axial direction of the capsule such that a relative axial position of the inner housing and the outer housing is configured to be varied so as to cause axial compression of the compressible seal.

8. The capsule according to claim 7, wherein the compressible seal has a cross-sectional height that is larger than a cross-sectional width thereof, and wherein the seal is provided in the groove defined in the inner housing.

9. The capsule according to claim 3, wherein a capillary height of the tubular fluid transfer element exceeds an axial height of a heating coil of the heater.

10. The capsule according to claim 3, wherein a heating coil of the heater has a height corresponding to 25%-50% of a height of the tubular fluid transfer element.

11. The capsule according to claim 10, wherein the tubular fluid transfer element has a capillary height corresponding to an actual height of the tubular fluid transfer element.

12. The capsule according to claim 3, wherein convection of the heater is between 4000 and 7000 W/m²K and power density is between 1.10 to 2.350 Watt/mm².

13. The capsule according to claim 12, wherein the power density is between 1.220 to 2.320 Watt/mm².

14. The capsule according to claim 12, wherein the power density is between 1.15 to 1.16 Watt/mm².

15. The capsule according to claim 3, wherein the at least one liquid inlet has a diameter of between 0.95 and 1.15 mm.

16. The capsule according to claim 3, wherein the at least one liquid inlet has a diameter of between 1.03 and 1.14 mm.

17. The capsule according to claim 3, wherein a heating coil of the heater has a height corresponding to 25%-45% of a height of the tubular fluid transfer element.

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18. The capsule according to claim **3**, wherein a heating coil of the heater has a height corresponding to 35% of a height of the tubular fluid transfer element.

19. The capsule according to claim **3**, wherein the protrusion is configured to extend into the groove at the variable depth along the axis. 5

20. The capsule according to claim **3**, wherein the groove is defined in an annular direction about the axis.

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